

features (for the most part) and work on protocols developed almost 30 years ago during the 1970s through the 1990s.

Addressing risks in ICSes at the field level

When you look at the field level in more depth, two primary points that present risk to the ICS are remote field sensors and I/O modules. At stake are uptime, predictable maintenance and overall industry efficiency—the cornerstones of the IIoT.

Risks with remote field sensors. Physical security of all sensors may not always be possible, especially when the sensors are very remote like those used to monitor oil and natural gas fields. Inaccessibility further makes these vulnerable to physical attacks, so it is essential to authenticate all sensors before their data is accepted. However, in most cases, field sensors, even those used in critical infrastructure systems, are both open and trusted. This vulnerability of the field sensors has not gone unnoticed.

In 2014, the well-known Black Hat conference featured a paper by Russian researchers who concluded that attacking an enterprise system directly is too much trouble. Instead, they concocted and described a 'Man in the Middle' attack by spoofing and replacing an open and trusted HART modem based field sensor. They have described the process in detail and even made the software libraries available for download.

Focus on secure systems for the IIoT must begin with a trusted sensor that is sending data to the Cloud or the PLC. Implications of a security breach on these remote devices can be profound.

Risks with I/O modules. Another way to get access to an open trusted system is by using cloned I/O modules to deliver malware. Factory owners are used to replacing I/O modules within their PLCs. There have been cases in Asia where cloned I/O modules (with a fake corporate logo of some of the top automation vendors) are available in the market. Again, since this is a trusted system with traditionally little to no embedded security, it can be an effective vehicle for delivering malware to the main PLC CPU. Physical security (making sure that PLC system updates are limited to a select set of people) can deter this, but keep in mind that this does not need to be a malicious act. You might not even know whether this is a cloned or a fake I/O module.

Implementing hardware authentication with a field sensor. A systems solution for the potential risks described can be simple. If you are going to trust the data from a slave sub-system, the data

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